在 CentOS Stream 9 上部署 Nginx

简介

Nginx(engine X)是一个高性能的 HTTP 和反向代理 Web 服务器，同时也提供了 ICMP/POP3/SMTP 服务。同 Tomcat 一样，Nginx 可以托管用户编写的 Web 程序成为可访问的页面服务，同时也可以作为流量代理服务器，控制流量的中转。

安装

Nginx 安装同样需要配置额外的 yum 仓库，才能使用 yum 安装

1.安装 yum 依赖程序

```
1  yum install -y yum-utils
```

```
[root@localhost ~]# yum install -y yum-utils
CentOS Stream 9 - BaseOS
```

等待安装完成

2.手动添加，nginx 的 yum 仓库（与前面部署 MySQL 不同）

```
[root@localhost ~]# cd /etc/yum.repos.d/
[root@localhost yum.repos.d]# pwd
/etc/yum.repos.d
[root@localhost yum.repos.d]# ll
总用量 44
-rw-r--r--. 1 root root 4245  4月   8 05:09 centos-addons.repo
-rw-r--r--. 1 root root 2600  4月   8 05:09 centos.repo
-rw-r--r--. 1 root root 1142  8月  17  2023 epel-cisco-openh264.repo
-rw-r--r--. 1 root root 1519  8月  17  2023 epel-next.repo
-rw-r--r--. 1 root root 1621  8月  17  2023 epel-next-testing.repo
-rw-r--r--. 1 root root 1453  8月  17  2023 epel.repo
-rw-r--r--. 1 root root 1552  8月  17  2023 epel-testing.repo
-rw-r--r--. 1 root root 1751  7月  26  2023 mysql-community-debuginfo.repo
-rw-r--r--. 1 root root 1478  7月  26  2023 mysql-community.repo
-rw-r--r--. 1 root root 1611  7月  26  2023 mysql-community-source.repo
[root@localhost yum.repos.d]#
```

可以看到本机的 yum 仓库，以.repo 结尾

在该文件夹下创建一个 nginx.repo，作为 yum 安装 nginx 的仓库配置文件

```
1 vim nginx.repo
```

写入配置文件（文件由 deepseek 生成，仅供参考）

```
[baseos]
name=CentOS Stream 9 - BaseOS - Aliyun
baseurl=https://mirrors.aliyun.com/centos-stream/9-stream/BaseOS/x86_64/os/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
enabled=1

[appstream]
name=CentOS Stream 9 - AppStream - Aliyun
baseurl=https://mirrors.aliyun.com/centos-stream/9-stream/AppStream/x86_64/os/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
enabled=1

[extras]
name=CentOS Stream 9 - Extras - Aliyun
baseurl=https://mirrors.aliyun.com/centos-stream/9-stream/Extras/x86_64/os/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
enabled=1

[crb]
name=CentOS Stream 9 - CRB - Aliyun
baseurl=https://mirrors.aliyun.com/centos-stream/9-stream/CRB/x86_64/os/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
enabled=1
~
```

3。通过 yum 指令安装稳定版的 Nginx

```
1  yum install -y nginx
```

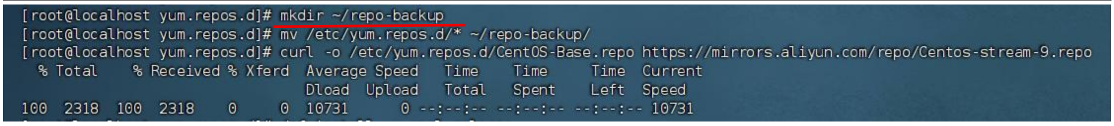
```
[root@localhost yum.repos.d]# yum install -y nginx
Repository baseos is listed more than once in the configuration
Repository appstream is listed more than once in the configuration
Repository crb is listed more than once in the configuration
CentOS Stream 9 - Extras - Aliyun                                    5.8 kB/s | 2.3 kB     00:00
Errors during downloading metadata for repository 'extras':
  - Status code: 404 for https://mirrors.aliyun.com/centos-stream/9-stream/Extras/x86_64/os/repodata/repomd.xml (IP: 60.169.2.220)
错误: 为仓库 'extras' 下载元数据失败 : Cannot download repomd.xml: Cannot download repodata/repomd.xml: All mirrors were tried
```

这次安装过程中出现报错，核心原因：/etc/yum.repos.d/ 目录里**仓库文件太多、严重冗余冲突**，这就是之前报「仓库重复、404 下载失败」的根源。

解决方法：

（1）．一键备份所有现有混乱源

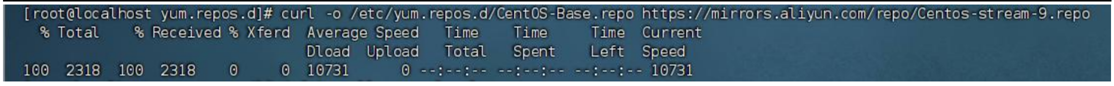
```
1 mkdir ~/repo-backup
```



```
[root@localhost yum.repos.d]# mkdir ~/repo-backup
[root@localhost yum.repos.d]# mv /etc/yum.repos.d/* ~/repo-backup/
[root@localhost yum.repos.d]# curl -o /etc/yum.repos.d/CentOS-Base.repo https://mirrors.aliyun.com/repo/Centos-stream-9.repo
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  2318  100  2318    0     0  10731      0 --:--:-- --:--:-- --:--:-- 10731
```

（2）．写入 CentOS Stream 9 可用的官方阿里源

```
1 curl -o /etc/yum.repos.d/CentOS-Base.repo https://mirrors.aliyun.com/repo/Centos-stream-9.repo
```

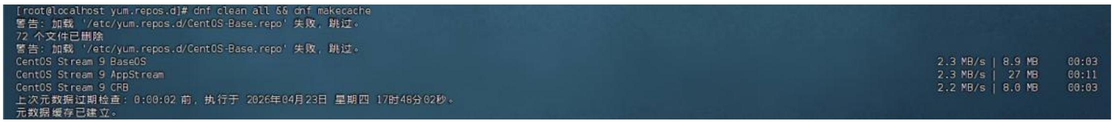


```
[root@localhost yum.repos.d]# curl -o /etc/yum.repos.d/CentOS-Base.repo https://mirrors.aliyun.com/repo/Centos-stream-9.repo
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  2318  100  2318    0     0  10731      0 --:--:-- --:--:-- --:--:-- 10731
```

（3）．写入 **100% 适配 Stream9、可正常访问**的阿里官方源（如果上面操作正常，跳过此步）

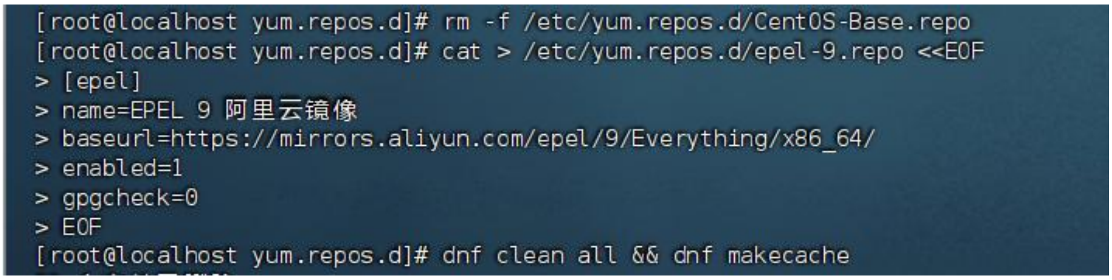
```
[root@localhost yum.repos.d]# cat > /etc/yum.repos.d/centos-stream9.repo <<EOF
> [baseos]
> name=CentOS Stream 9 BaseOS
> baseurl=https://mirrors.aliyun.com/centos-stream/9-stream/BaseOS/x86_64/os/
> enabled=1
> gpgcheck=0
>
> [appstream]
> name=CentOS Stream 9 AppStream
> baseurl=https://mirrors.aliyun.com/centos-stream/9-stream/AppStream/x86_64/os/
> enabled=1
> gpgcheck=0
>
> [crb]
> name=CentOS Stream 9 CRB
> baseurl=https://mirrors.aliyun.com/centos-stream/9-stream/CRB/x86_64/os/
> enabled=1
> gpgcheck=0
> EOF
```

（4）．重建缓存，验证源可用



```
[root@localhost yum.repos.d]# dnf clean all && dnf makecache
警告: 加载 '/etc/yum.repos.d/CentOS-Base.repo' 失败，跳过。
72 个文件已删除
警告: 加载 '/etc/yum.repos.d/CentOS-Base.repo' 失败，跳过。
CentOS Stream 9 BaseOS                               2.3 MB/s | 8.9 MB     00:03
CentOS Stream 9 AppStream                            2.3 MB/s |  27 MB     00:11
CentOS Stream 9 CRB                                  2.2 MB/s | 8.0 MB     00:03
上次元数据过期检查: 0:00:02 前，执行于 2026年04月23日 星期四 17时48分02秒。
元数据缓存已建立。
```

（5）．手动创建 EPEL 阿里镜像（彻底规避国外源超时），并且之后再次更新缓存，等待完成



```
[root@localhost yum.repos.d]# rm -f /etc/yum.repos.d/CentOS-Base.repo
[root@localhost yum.repos.d]# cat > /etc/yum.repos.d/epel-9.repo <<EOF
> [epel]
> name=EPEL 9 阿里云镜像
> baseurl=https://mirrors.aliyun.com/epel/9/Everything/x86_64/
> enabled=1
> gpgcheck=0
> EOF
[root@localhost yum.repos.d]# dnf clean all && dnf makecache
```

（6）.完成以上步骤，手动使用 yum 指令，安装 Nginx

```
[root@localhost yum.repos.d]# yum install -y nginx
上次元数据过期检查：0:00:40 前，执行于 2026年04月23日 星期四 17时52分52秒。
依赖关系解决。
```

安装完成，安装版本为 Nginx-1.20.1

```
[root@localhost ~]# nginx -version
nginx version: nginx/1.20.1
[root@localhost ~]#
```

Nginx-1.20.1 是一个比较老的版本，但用于文档练习，本地测试完全够用，如果要用在正式项目或者内网服务，建议使用更新以及更稳定的 1.24.x/1.26.x 版本

4.启动 Nginx

```
1  systemctl start nginx
2  systemctl status nginx
```

启动 nginx.并查看 nginx 的状态，此时处于 active(启动状态)

```
[root@localhost ~]# systemctl start nginx
[root@localhost ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-04-23 18:22:51 CST; 1s ago
    Process: 340083 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 340084 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 340085 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 340086 (nginx)
      Tasks: 5 (limit: 22966)
     Memory: 4.5M (peak: 4.8M)
        CPU: 43ms
     CGroup: /system.slice/nginx.service
             ├─340086 "nginx: master process /usr/sbin/nginx"
             ├─340087 "nginx: worker process"
             ├─340088 "nginx: worker process"
             ├─340089 "nginx: worker process"
             └─340090 "nginx: worker process"

4月 23 18:22:50 localhost.localdomain systemd[1]: Starting The nginx HTTP and reverse proxy server...
4月 23 18:22:50 localhost.localdomain nginx[340084]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
4月 23 18:22:50 localhost.localdomain nginx[340084]: nginx: configuration file /etc/nginx/nginx.conf test is successful
4月 23 18:22:51 localhost.localdomain systemd[1]: Started The nginx HTTP and reverse proxy server.
[root@localhost ~]#
```

5.配置防火墙放行

Nginx 默认绑定 80 端口，需要关闭防火墙护或者放行 80 端口

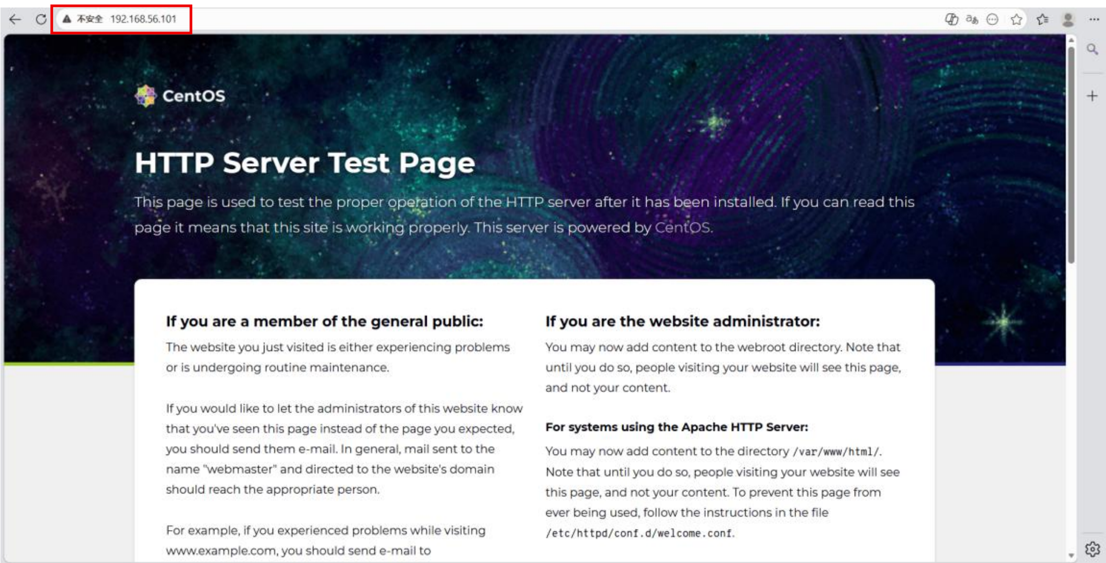
这里以关闭防火墙为例（平时测试或者实验都可以选择关闭防火墙）

```
1 systemctl stop firewalld   //关闭防火墙
2 systemctl stop firewalld   //查看防火墙是否关闭成功
```

```
[root@localhost ~]# systemctl stop firewalld
[root@localhost ~]# systemctl status firewalld
○ firewalld.service - firewalld - dynamic firewall daemon
     Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
     Active: inactive (dead) since Thu 2026-04-23 18:31:23 CST; 2s ago
   Duration: 1h 41min 54.906s
       Docs: man:firewalld(1)
    Process: 863 ExecStart=/usr/sbin/firewalld --nofork --nopid $FIREWALLD_ARGS (code=exited, status=0/SUCCESS)
   Main PID: 863 (code=exited, status=0/SUCCESS)
        CPU: 865ms

4月 23 16:49:27 localhost systemd[1]: Starting firewalld - dynamic firewall daemon...
4月 23 16:49:28 localhost systemd[1]: Started firewalld - dynamic firewall daemon.
4月 23 18:31:23 localhost.localdomain systemd[1]: Stopping firewalld - dynamic firewall daemon...
4月 23 18:31:23 localhost.localdomain systemd[1]: firewalld.service: Deactivated successfully.
4月 23 18:31:23 localhost.localdomain systemd[1]: Stopped firewalld - dynamic firewall daemon.
[root@localhost ~]#
```

6.关闭防火墙后，在浏览器输入 Linux 服务器的 IP 地址或者主机名即可访问



成功访问 Nginx